



# Future climate risk and urban tree inventories in Australian cities: Pitfalls, possibilities and practical considerations

Manuel Esperon-Rodriguez<sup>\*</sup>, Sally A. Power, Mark G. Tjoelker, Paul D. Rymer

Hawkesbury Institute for the Environment, Western Sydney University, Locked Bag1797, Penrith, NSW 2751 Australia

## ARTICLE INFO

Handling Editor: Dr. Cecil Konijnendijk van den Bosch

### Keywords:

Climate change  
Survey  
Urban greening  
Urban management  
Urban planning

## ABSTRACT

Urban tree inventories are useful tools to assess the environmental and socio-economic services provided by urban forests. These inventories enable the evaluation of the climate change risk to urban forests, and governments rely on such inventories for urban planning and management. Here, we assessed the future climate risk of Australia and the state of urban tree inventories across 116 local government areas (LGAs), representing 21 % of the country's LGAs and encompassing 55 % of the national human population. We evaluated projected changes in temperature and precipitation by 2050 for each LGA and conducted a survey to obtain information on the extent and types of data available in existing urban tree inventories. Additionally, we compiled demographic, socio-economic, and geographical data for all LGAs to explore correlates with tree inventory status. Temperature increases in 2050 were predicted in all LGAs, with higher latitude and smaller LGAs identified to undergo greater increases in temperature compared to larger and lower latitude LGAs. Decreases in seasonal precipitation were predicted for 97 LGAs. Seventy-six (66 %) of surveyed LGAs had urban tree inventories, which most commonly included trees along streets and in parks. Sixty-one LGAs record information on tree mortality, while 31 LGAs dynamically update their inventories. The presence of an inventory and the area it covered were positively associated with human population density. More than 30 years ago, in 1988, John Gray wrote that "insufficient statistics were available in Australia to provide an accurate picture of the urban forest estate". Our research shows there has not been a significant advance in the adoption and use of urban forest inventories over the past three decades. Long-term, dynamically updated inventories are crucial for urban forest management to inform planting choices to support sustainable and resilient cities.

## 1. Introduction

The pace at which climate is changing poses a serious threat to the continued persistence of urban forests globally (Esperon-Rodriguez et al., 2022a; IPCC, 2021). While the impacts of climate change on natural ecosystems, both terrestrial and aquatic, have been widely studied (e.g. Hoegh-Guldberg and Bruno, 2010; Lenoir et al., 2020; Scholze et al., 2006), less is known about the potential impacts on urban ecosystems (but see Brandt et al., 2016; Esperon-Rodriguez et al., 2022a; Ordóñez and Duinker, 2014). Urban forests are recognised for multiple environmental and socio-economic services they provide to citizens globally, such as increasing biodiversity and reducing air pollutants (García-Sánchez et al., 2019; Livesley et al., 2016). These forests can also mitigate the effects of climate change via carbon sequestration and reduction of heat by altering microclimatic conditions (Akbari et al., 2001). The provision of these services, however, is compromised by

global climate change, extreme climatic conditions (e.g. heatwaves, drought, extreme storms), pests and diseases, and inadequate management practices, all of which contribute to increased tree mortality (Brandt et al., 2016; Esperon-Rodriguez et al., 2022b; Hilbert et al., 2019; Marchin et al., 2022; Ordóñez and Duinker, 2014).

Urban greening policies are aimed at the strategic delivery of ecosystem services and benefits (Young and McPherson, 2013). However, for these services to be provided efficiently over the coming decades, it is imperative to assess the exposure to climate change in cities and identify tree species that are most vulnerable in order to evaluate the risk of poor growth and failure under present and future climates (Esperon-Rodriguez et al., 2022b). Urban forest managers often seek to optimise the use of resources and minimise losses when investing in urban forestry programs (Hilbert et al., 2019). Therefore, successful urban greening involving increases in tree canopy cover for the future must be planned to minimise losses expected due to climate change.

<sup>\*</sup> Corresponding author.

E-mail address: [m.esperon-rodriguez@westernsydney.edu.au](mailto:m.esperon-rodriguez@westernsydney.edu.au) (M. Esperon-Rodriguez).

<https://doi.org/10.1016/j.ufug.2022.127769>

Received 8 April 2022; Received in revised form 20 October 2022; Accepted 21 October 2022

Available online 26 October 2022

1618-8667/© 2022 Elsevier GmbH. All rights reserved.

Presently, there is a lack of studies identifying climate change as a driver of urban tree dieback and mortality; however, urban tree inventories have been proposed as tools to aid the evaluation of climate change impacts on cities (Esperon-Rodriguez et al., 2022b). Urban tree inventories (hereafter tree inventories) are powerful management and planning tools that collect accurate information on individual trees planted in cities (Nielsen et al., 2014). These inventories can aid in the management and planning of urban trees (Miller et al., 2015; Morgenroth et al., 2016; Roman et al., 2020), which is crucial to assess tree mortality, growth and performance and inform decision-making in urban greening programs and species selection (Roman et al., 2020; Roman et al., 2015). Recently, tree inventories have also been used to identify causes of tree failure and species more prone to fail (Roman et al., 2022).

In recent years, the value of tree inventories for evaluating the performance of urban forests has been increasingly recognised by local governments and researchers (Ma et al., 2021; Nielsen et al., 2014; Vogt et al., 2017). In most cases, individual trees are recorded (including scientific name and location) and data on species' growth and health can also be collected (Nielsen et al., 2014; Nowak and Greenfield, 2018). These attributes (e.g. species, location and planting date) represent the basis of long-term monitoring (Roman et al., 2020) and when data are collected dynamically with the inclusion of growth metrics (e.g. height, diameter), they can inform urban management decisions (Alvarez et al., 2015; Nielsen et al., 2014; Östberg et al., 2018). When available, local governments rely on tree inventories for planning the investment needs of cities, such as greening practices by, for example, determining species richness and abundance (Alvarez et al., 2015; Nielsen et al., 2014; Östberg et al., 2018). Furthermore, tree inventories can also be used to inform the development of future, climate-proof and less vulnerable urban forests (Esperon-Rodriguez et al., 2022a).

Australia is experiencing more extreme weather than ever before, with record-breaking temperatures frequently reported over recent years (BM, 2018; BOM and CSIRO, 2020; Páscoa et al., 2022). As climate changes and temperatures continue to rise, the frequency and magnitude of heatwaves are predicted to increase (IPCC, 2021), with the potential for large, negative effects on urban forests (Esperon-Rodriguez et al., 2022a; Khan and Conway, 2020; Sanusi and Livesley, 2020). A large proportion of the tree species planted in Australian cities already experience climate conditions that exceed their physiological tolerance and the climate risk posed by warmer and drier conditions will be exacerbated in the future (Esperon-Rodriguez et al., 2022a; Esperon-Rodriguez et al., 2019; Zhang and Brack, 2021). Therefore, cities will need resilient urban forests capable of surviving increasingly extreme climates, to improve urban liveability and the well-being of city dwellers.

The impacts of climate change on cities, however, will be different depending on the geographic location of each city (e.g., latitude), the composition of the surrounding landscape matrix, and the relative proportion of hard surfaces and extent of green and blue infrastructure (Almaaitah et al., 2021; Chapman et al., 2017; Demuzere et al., 2014; Manoli et al., 2019). For example, cities where non-urban areas are dominant have different climates due to the vegetated land (and/or water bodies) land surrounding them. The capacity of cities to cope with the impacts of climate change will depend on the resources available in each city and their capacity to mitigate those changes. Here, we hypothesised that cities with more economic resources will have more tools to face the challenges of climate change, including the development of urban tree inventories. Indeed, urban tree inventories can reflect the demographic and socio-economic status of the cities where they are developed. Therefore, large, wealthy Australian cities are more likely to have inventories than less wealthy, small towns. This is the case for Sweden, where the number of municipalities with inventories was positively related to population size (Östberg et al., 2018) or in Italy, where the Italian Law 10/2013, "Rules for the development of urban green spaces", stipulates that all municipalities with > 15,000 residents

are required to develop an inventory of trees in public green areas (Italian Ministry of Ecological Transition L. 10/2013.).

This work aimed to assess the future climate risk of Australian cities in relation to the state of their current urban tree inventories. Our objectives were: (1) to determine the exposure of Australian cities to future climate change based upon modelled projections; (2) to identify which cities have urban tree inventories and assess the state, management and attributes collected in their inventories; and (3) to evaluate relationships between climate variables and socio-economic demographics with the existence of tree inventories and the scope of attributes collected. We address these objectives in the context of climate change risk in local government areas (LGA) and their associated municipalities throughout Australia.

## 2. Methods

### 2.1. Study area

Australia has a population of ~25.7 million people (Australian Bureau of Statistics 2020; <www.abs.gov.au>), with 90 % of the population living in 0.22 % of the country's land area (i.e. urban areas) and 71 % living in major cities (Cresswell and Murphy, 2017). Australia is divided into eight states and territories (Australian Capital Territory - ACT; New South Wales - NSW; Northern Territories - NT; Queensland - QLD; South Australia - SA; Tasmania - TAS; Victoria - VIC; and Western Australia - WA), each of which is divided into local government areas (LGAs), totalling 562 across Australia. LGAs comprise cities, towns, municipalities, regions, shires and districts that are managed as a unit to the extent permitted by local legislation (ASGS, 2011).

### 2.2. Exposure to climate change

To assess the exposure of each LGA to climate change, we obtained climate data (interpolations of observed data, representative of 1970–2000, i.e. baseline) from WorldClim version 2.1 (Fick and Hijmans, 2017) at 30 arc-seconds (~1 km) resolution. We selected four variables describing seasonal and extreme climate: maximum temperature of the warmest month (MTWM), minimum temperature of the coldest month (MTCM), precipitation of the driest quarter (PDQ) and precipitation of the warmest quarter (PWQ). For future climate data, we downloaded three Global Circulation Models (GCMs) that include CMIP6 downscaled future climate projections (BCC-CSM2-MR, China; CNRM-CM6-1, France; MIROC6, Japan) to estimate climate projections for 2050 (average for 2041–2060), considering the highest emissions of the Shared Socio-economic Pathways scenario (SSP585). By selecting multiple GCMs, we aimed to address the uncertainty and variability around future climate predictions. We obtained climate data from each GCM and used median values across all GCMs for subsequent analyses. All climate data were transformed to the Australian Albers Equal-Area Conic Projection (EPSG:3577) at ~1 km resolution.

LGA polygons were obtained as a shapefile (GDA94) from the Australian Statistical Geography Standard (ASGS) Volume 3 - Non-ABS Structures (cat no. 1270.0.55.003) (Australian Bureau of Statistics (<www.abs.gov.au>)). These data were projected to EPSG:3577 as above. For each LGA polygon, we placed a grid (1 × 1 km) over its area and extracted the values of all four climate variables at each cell and then, we calculated thresholds of the 95th percentile for MTWM and the 5th percentile of MTCM, PDQ and PWQ. We selected these thresholds to assess the climate extremes of these variables for each LGA. Throughout the manuscript, when we refer to these climate variables, we imply the use of the 95th (MTWM) and 5th (MTCM, PDQ and PWQ) percentiles, accordingly.

Exposure (*E*) is the degree to which a city is exposed to projected future climatic change (IPCC, 2001) and indicates how much the future climate differs (e.g. warmer or drier) from current baseline climatic conditions. Exposure was calculated as the difference between future

and baseline climate of each LGA as follows:

$$E = \text{LGA}_{\text{FutureClimate}} - \text{LGA}_{\text{BaselineClimate}} \quad (1)$$

A positive exposure ( $E > 0$ ) indicates that warmer or wetter conditions are expected under future climate change scenarios.

We acknowledge some caveats for our approach. First, climate data downscaling based on coarse-grained spatial interpolations may fail to identify local areas where climate extremes can be exacerbated by the UHI effect or moderated by more benign or cooler conditions caused by blue and green infrastructure (Alavipanah et al., 2015; Corburn, 2009; Kidder and Essenswanger, 1995). Second, larger LGAs might have higher spatial variability in climate extremes compared to smaller LGAs where climatic conditions can be more uniform. Finally, our determination of future climate change exposure does not account for the effects of future urban population growth and urban land use changes, which could further intensify climate change, suggesting that our estimates are conservative.

### 2.3. Australian urban tree inventories survey

To assess the state of urban tree inventories in Australian cities, we contacted LGAs with populations  $> 10,000$  people (based on the 2018 population census) — with the addition of Yankalilla, South Australia (5679 people) - resulting in a total of 208 LGAs (37 %) contacted across all eight Australian states (Table 1). Personnel employed in positions related to urban planning and forestry at each LGA were contacted individually and asked to respond to a written survey. The survey was available from September to December 2019. The online survey had 28 questions divided into five sections: (1) general information about the LGA and whether or not they have a tree inventory; (2) details on inventory data, including questions about the types of data collected, the number of trees in the inventory and dates of tree establishment; (3) tree attributes and monitoring details, with questions covering the establishment period (i.e. new plantings,  $< 2$  years) as well as established trees (i.e. old plantings,  $> 2$  years); (4) changes in tree status, with questions about data updates, records of failures and mortality; and (5) public availability of inventory data and information, including questions on urban forest planning strategies (Appendix A).

### 2.4. Demographic and socio-economic data

For each LGA, we collected data for total population size (hereafter population) and population density (data for 2020; Australian Bureau of Statistics <www.abs.gov.au>) and four socio-economic indices that rank areas in Australia according to relative socio-economic advantage and disadvantage: (1) Index of Economic Resources (IER), which focuses on the financial aspects of relative socio-economic advantage and disadvantage, by summarising variables related to income and wealth; (2) Index of Education and Occupation (IEO), designed to reflect the educational and occupational level of communities; (3) Index of Relative Socio-economic Disadvantage (IRSeD), which is a general socio-

economic index that summarises the economic and social conditions of people and households within an area; and (4) Index of Relative Socio-economic Advantage and Disadvantage (IRSeAD), which summarises the economic and social conditions of people and households within an area, including both relative advantage and disadvantage measures. The indices are based on information from the five-yearly Census of Population and Housing 2016 (ABS 2016).

### 2.5. Data analysis

We developed a Principal Component Analysis (PCA) to identify the most important demographic (i.e. population, population density), socio-economic (i.e. four socio-economic indices), climate (i.e. exposure [change] to MTWM, MTCM, PDQ and PWQ) and geographical (i.e. area, latitude, longitude) variables in our dataset. We calculated the relative importance of each component by measuring the length of each vector (Legendre and Legendre, 2012). We identified the principal sources of variability and removed correlated variables. Finally, we selected for subsequent analyses the exposure to MTWM and PDQ, IEO, population size and density, and LGA area (Supplemental Fig. S1).

We used generalised linear models to test the hypotheses: (1) the presence of tree inventory and the number of inventoried trees increases with population size and density and LGA area; and (2) the number of attributes collected in tree inventories decreases with the number of trees inventoried within an LGA. Additionally, we assessed relationships between the exposure to climate change at each LGA with demographic and socio-economic variables. Having a tree inventory was modelled as a binomial response using the “*genmod*” procedure with a binomial distribution and “logit link” function (i.e. binary logistic regression). As explanatory variables, we included: population size and density, area and IEO. LGA name was included as random effect and model statistical significance was assessed using Tukey’s test (Legendre and Legendre, 2012). Logistic regressions were used to test relationships between the extent of tree inventory (i.e. percentage of total ground area included in the tree inventory) and number of trees in each LGA with population size and density, area, IEO, exposure to MTWM and PDQ. Logistic regressions were also used to test the relationship between the number of attributes collected in tree inventories and the number of trees at each LGA. The independent and interactive effects between predicted climate change exposure for temperature (exposure to MTWM) and precipitation (exposure to PDQ) were evaluated using linear mixed models with the “*lm*” function (Chambers and Hastie, 1992). Finally, we assessed the relationship between exposure to MTWM and PDQ with LGA latitude and longitude using linear mixed models to assess geographic patterns. Model performance was evaluated through the calculation of an  $R^2$  value and the  $F$ -statistic at a significance level of  $P < 0.05$ . All statistical analyses were conducted using R version 4.0.5 (R Core Team, 2021), specifically the package *Vegan* (Oksanen et al., 2019).

## 3. Results

### 3.1. Exposure to climate change

All LGAs were predicted to have increases in temperature for both MTWM and MTCM. Ninety-seven of the 116 LGAs were predicted to have decreases in PDQ, while 85 LGAs were predicted to have decreases of PWQ (Fig. 1). On average and across all LGAs, MTWM and MTCM were predicted to increase  $2.1^\circ\text{C}$  ( $\pm 0.2^\circ\text{C}$ ) and  $1.4^\circ\text{C}$  ( $\pm 0.2^\circ\text{C}$ ), respectively, by 2050. As for precipitation, PDQ and PWQ were predicted to decrease on average by  $-8.0\text{ mm}$  ( $\pm 7.3\text{ mm}$ ) and  $-7.4\text{ mm}$  ( $\pm 13.2\text{ mm}$ ), respectively (Table 2; Supplemental Table S1).

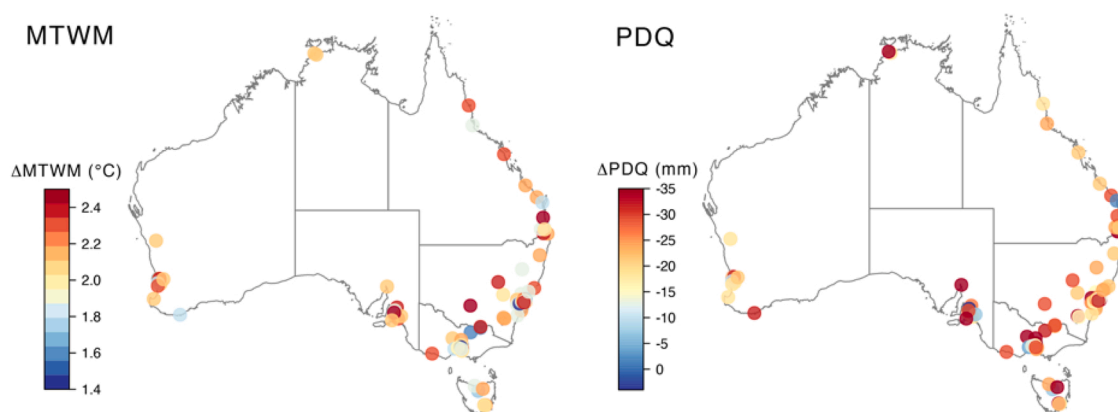
### 3.2. Australian urban tree inventories survey

We received 116 survey responses (56 %) from the 208 LGAs contacted (Table 1). Of these 116 LGAs, 76 (66 %) reported having a tree

**Table 1**

The total number of local government areas (LGAs) contacted across eight Australian states and territories, number of LGA survey respondents (Responses) and number and proportion (in brackets) of respondent LGAs with urban tree inventories (Inventory).

| State                              | Total | Responses | Inventory |
|------------------------------------|-------|-----------|-----------|
| Australian Capital Territory (ACT) | 2     | 1         | 1 (100 %) |
| New South Wales (NSW)              | 60    | 42        | 22 (52 %) |
| Northern Territories (NT)          | 3     | 2         | 1 (50 %)  |
| Queensland (QLD)                   | 21    | 12        | 9 (75 %)  |
| South Australia (SA)               | 23    | 13        | 9 (69 %)  |
| Tasmania (TAS)                     | 13    | 6         | 5 (83 %)  |
| Victoria (VIC)                     | 44    | 19        | 15 (79 %) |
| Western Australia (WA)             | 42    | 19        | 14 (74 %) |



**Fig. 1.** Predicted climate change exposure to changes in maximum temperature of the warmest month (MTWM) and precipitation of the driest quarter (PDQ) by 2050 under the SSP585 scenario for 116 local government areas in Australia. For details see [Supplemental Table S1](#).

**Table 2**

Average, standard deviation (SD), maximum (Max) and minimum (Min) values across 116 local government areas (LGAs) for baseline (interpolations of observed data, representative of 1970–2000) and future 2050 climate for maximum temperature of the warmest month (MTWM; °C), minimum temperature of the coldest month (MTCM; °C), precipitation of the driest quarter (PDQ; mm) and precipitation of the warmest quarter (PWQ; mm). Names in brackets identify LGAs with max and min values and their states. For states abbreviations see [Table 1](#). Data from WorldClim version 2.1 ([Fick and Hijmans, 2017](#)).

| Variable      | Average | SD    | Max                             | Min                             |
|---------------|---------|-------|---------------------------------|---------------------------------|
| Baseline MTWM | 28.5    | 2.7   | 37.3<br>(Greater Geraldton, WA) | 21.8<br>(Devonport, TAS)        |
| Future MTWM   | 30.6    | 2.7   | 39.2<br>(Greater Geraldton, WA) | 23.6<br>(Glenorchy, TAS)        |
| Baseline MTCM | 6.7     | 3.2   | 18.4<br>(Darwin, NT)            | -0.2<br>(Canberra, ACT)         |
| Future MTCM   | 8.1     | 3.2   | 20.2<br>(Darwin, NT)            | 1.1<br>(Canberra, ACT)          |
| Baseline PDQ  | 112.7   | 57.2  | 235.6<br>(Mosman, NSW)          | 4<br>(Litchfield, NT)           |
| Future PDQ    | 104.7   | 53.2  | 219.4<br>(Mosman, NSW)          | 4<br>(Litchfield, NT)           |
| Baseline PWQ  | 208.1   | 158.1 | 1002.4<br>(Cairns, QLD)         | 31.1<br>(Greater Geraldton, WA) |
| Future PWQ    | 200.6   | 156.0 | 1057.8<br>(Cairns, QLD)         | 34<br>(Greater Geraldton, WA)   |

inventory. Of the 76 LGAs with inventories, 93 % (71 LGAs) reported keeping records of individual trees, which are identified to the species level with details of their location (e.g. latitude/longitude, map grid, address). Twenty-nine LGAs (38 %) reported having between 50 % and 90 % of their LGA area included in their tree inventories. While having < 10,000 trees was the most frequent answer for the number of trees included in the inventory (21 LGAs, 28 %), 18 LGAs (24 %) reported keeping records for 50,000–100,000 individual trees, and seven LGAs (9 %) included > 100,000 trees in their inventories. Only one LGA (Clarence Valley Council, NSW) established its inventory before 1989, while 27 LGAs (36 %) established their inventories after 2015. Six LGAs (8 %) have static inventories (i.e. data only collected once after the establishment), but 31 LGAs (41 %) update their inventories dynamically (e.g. adding/removing records as trees are planted/lost) ([Fig. 2](#)). In most LGAs (83 %), inventories are developed by in-house arborists (42 %), external arborists (32 %) or both (9 %), whereas the other LGAs use external contractors.

New plantings (i.e. < two years) are incorporated into the inventories after planting in 56 LGAs (74 %) and these are monitored

during the first month after establishment in 31 LGAs (41 %). Species name, location and planting date are the most common attributes collected after tree establishment. Established trees (i.e. > three years) are monitored every five years in 27 LGAs (36 %). New plantings are integrated into the inventories immediately after planting in 47 LGAs (63 %) and after an establishment period (> two years) in nine LGAs (12 %). Eight LGAs (11 %) reported not integrating new plantings into their inventories ([Fig. 2](#)).

Attributes are mainly collected by in-house arborists (50 % LGAs), contractors (14 %) or external arborists (5 %), whereas the rest of the LGAs use technical officers or other personnel. The most common attributes collected are species' scientific name (71 LGAs, 93 %), location (62 LGAs, 82 %) and planting date (60 LGAs, 79 %). Only one LGA reported collecting iTree eco-attributes and photos (Cairns, QLD). The average number of attributes collected by each LGA was seven (standard deviation  $\pm 4$ ), and LGAs with 20,000–50,000 and > 100,000 trees collected the most attributes with an average of eight ( $\pm 4$ ). Almost all LGAs (97 %) reported updating information with management/operational actions, where tree removal (86 %), tree health (70 %), pruning (59 %), remedial work (59 %) and watering (50 %) were the most common attributes updated.

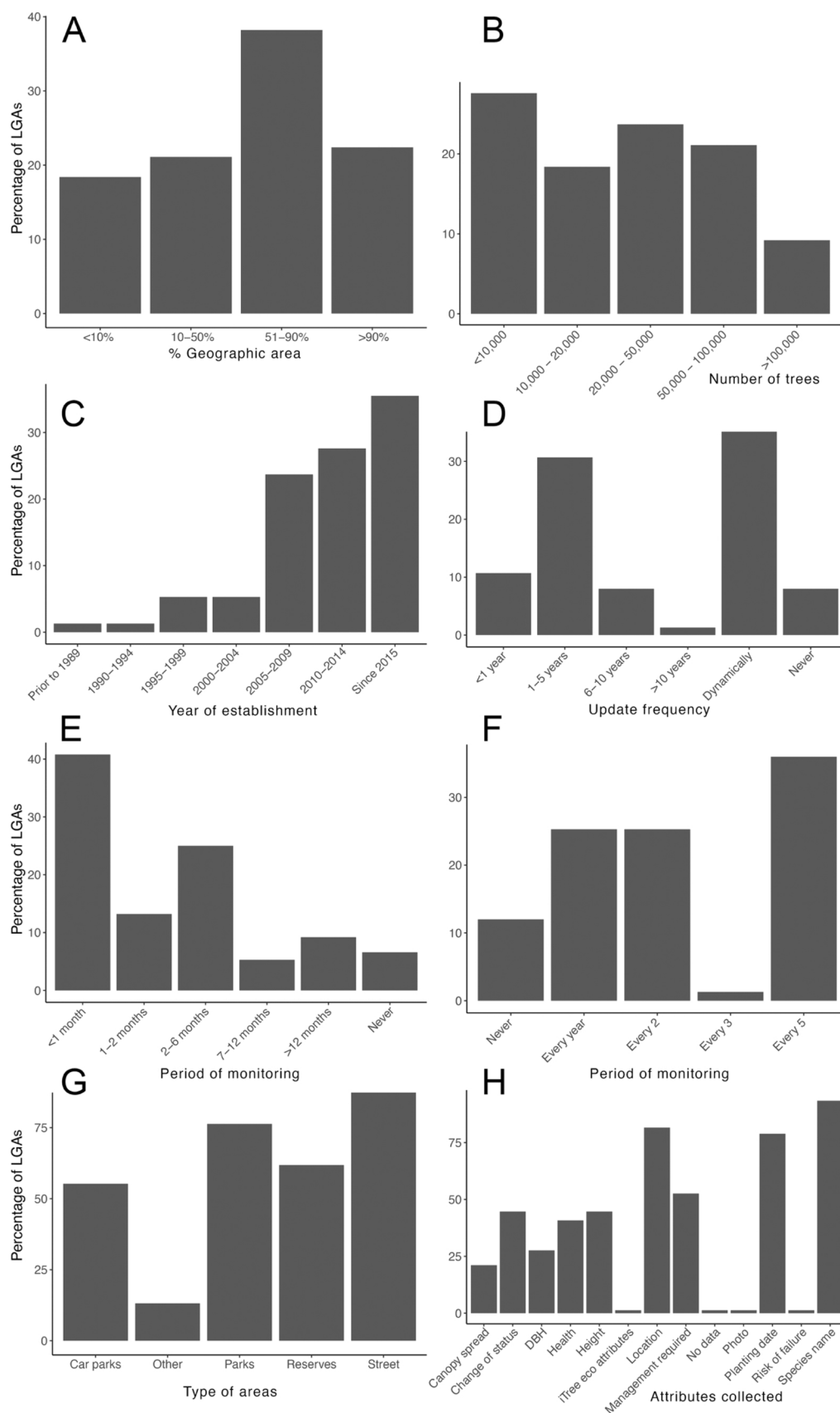
Thirty-two LGAs (42 %) record changes in health status of trees (e.g. leaf browning/drop), 61 LGAs (80 %) record tree mortality and 47 LGAs (62 %) record tree failures. The most frequent causes of failure reported are vandalism (59 LGAs, 78 %), storm action (53 LGAs, 70 %), urban development (44 LGAs, 58 %), pest and diseases (43 LGAs, 57 %), car incidents (31 LGAs, 41 %), drought or heatwave events (31 LGAs, 41 %), and poor planting (21 LGAs, 28 %). Climate change was attributed as the primary cause of tree failure in only seven LGAs (9 %).

Fifty-five LGAs (72 %) incorporate their inventories into the LGA mapping system (i.e. the GIS/mapping framework), which facilitates planning and management. Regarding data sharing, tree inventories of seven LGAs (9 %) are available through their respective websites, 31 LGAs (41 %) inventories are available on request, and 26 LGAs (37 %) do not provide inventory data. Finally, 41 LGAs (50 %) have an urban forest planning strategy, whereas for 23 LGAs (30 %) these strategies are under development, and 13 LGAs (20 %) have no urban forest strategies.

### 3.3. Demographic and socio-economic data

Across the full LGA dataset, the median population size was 74,800 people ( $\pm 150,901$ ) and the median population density was 513 people/km<sup>2</sup> ( $\pm 1842$ ). The median area across all LGAs was 271 km<sup>2</sup> ( $\pm 2,474$ ), with the largest and smallest areas in Gladstone (QLD, 10,484 km<sup>2</sup>) and Claremont (WA, 5 km<sup>2</sup>), respectively. On average, ACT (IRSeD and IRSeAD) and NSW (IEO and IER) had the highest values for socio-economic indices, while NT (IER) and Tasmania (IEO, IRSeD and





**Fig. 2.** Attributes of urban tree inventories in Australia. Percentages of respondent local government areas (LGAs) with the following tree inventory attributes: (A) geographic area of LGA in inventory (%), (B) number of trees, (C) year of inventory establishment, (D) update frequency, (E) time period of monitoring new plantings (< 2 years), (F) time period of monitoring established plantings (>3 years), (G) types of areas included, and (H) attributes collected in urban tree inventories from 76 local government areas (LGAs) in Australia.

IRSeAD) had the lowest. Nonetheless, we found no patterns when we assessed the four indices geographically (Supplemental Table S2).

### 3.4. Relationship among climatic, demographic, and socio-economic variables

The probability of LGAs having a tree inventory increased with increasing LGA area and population density. The areal extent (i.e. proportion of total geographic area) of LGAs included in inventories was correlated with population density and IEO. No correlations were found between number of trees and population size, population density, LGA area, and IEO. We found little evidence that LGAs with a higher number of trees collected fewer attributes in their inventories. We found a correlation between LGA area and population density with IEO (Table 3).

We found evidence that larger LGAs (i.e. area and population size) are at most risk of climate change (i.e. exposure) compared to smaller LGAs. Future changes in MTWM were significantly correlated to IEO and LGA area; that is, smaller LGAs were predicted to have higher increases in MTWM. The greatest increases of MTWM were associated with LGAs with inventories. Although the changes in PDQ were small (ranging from 3.4 mm [2 %] in Orange, NSW, to −32 mm [−18 %] in Strathfield, NSW), PDQ exposure was significantly correlated with LGA area, population size and population density. More populated LGAs with larger areas were predicted to have greater decreases in PDQ compared to less populated and smaller LGAs. This finding indicates that more people can be affected by decreases in precipitation and potentially, larger areas might require additional irrigation or altered species selection to offset the impacts of low PDQ. The greatest decreases in PDQ were associated with LGAs with inventories (Table 3).

Finally, we found a significant correlation between predicted changes in MTWM and PDQ, where LGAs predicted to undergo higher increases in MTWM were also associated with greater decreases in PDQ. Also, we found significant correlations between predicted changes of MTWM and latitude and between predicted changes of PDQ and

longitude. That is, LGAs at more southerly high latitudes were predicted to undergo higher increases in temperature, and LGAs at low longitude (towards the west, i.e. inland) are predicted to have greater reductions in PDQ (Table 3).

## 4. Discussion

Future climate change will have adverse consequences for city dwellers and governments in Australian cities, although the magnitude of these consequences will vary across cities. All LGAs assessed here were predicted to become warmer in 2050 and the majority were predicted to undergo decreases in seasonal precipitation (i.e. PDQ and PWQ) by 2050. These conditions will impact urban forests, and in many cases, these forests will require management interventions to thrive (Esperon-Rodriguez et al., 2022a; Teodorescu, 2010).

Urban forests in cities predicted to have high increases in temperature and reductions in precipitation are more vulnerable to climate change (Esperon-Rodriguez et al., 2022a). Previous research showed that, although species' vulnerability to climate change varies among cities due to differences in species' climatic tolerance and management practices, the impact of climate change on urban trees can have negative socio-economic effects (Ramage et al., 2013; Roman et al., 2014). We found evidence that LGAs of larger geographic area, population size and density are projected to have slightly greater reductions in rainfall (PDQ) by 2050, compared to smaller, less densely populated LGAs. This finding implies that trees in larger more densely populated LGAs will potentially require human intervention during times of water deficit, implying greater management costs. However, large LGAs with high population densities tend to have high socio-economic indices (i.e. IEO), which suggests that these LGAs might be capable of developing measures to mitigate future climate change (i.e. PDQ exposure) through management actions, such as stormwater capture and irrigation. Irrigation, nonetheless, may not be a sustainable solution in many places where water is increasingly scarce (Di Baldassarre et al., 2018; Hoekstra and Mekonnen, 2012).

Australia's population is dominated by coastal towns and cities, with 85 % of the population living within 50 kilometres of the coast (Cresswell and Murphy, 2017). Australian LGAs that are inland or regional are dominated by non-urban forests or agricultural land. Therefore, the buffering effect of vegetation (e.g. high canopy cover) surrounding these LGAs can alter the climatic conditions in those urban areas (Alavipanah et al., 2015). In coastal cities, cloud conditions and wind speed can also alter microclimate and mitigate urban heat (Kidder and Essenwanger, 1995).

We found that the greatest increases in MTWM were in smaller and more southerly LGAs, where the potential cooling benefit provided by urban trees may play an important role. However, predicted warm conditions can also increase the risk of tree dieback and mortality (Allen et al., 2010; Marchin et al., 2022). Furthermore, more populous and inland LGAs are predicted to have the greatest decreases in PDQ, suggesting these large, urban populations will be at more risk of climate change than those in smaller cities. Undoubtedly, these conditions will have socio-economic and human health impacts (Haines et al., 2006; Hunt and Watkiss, 2011).

In Sweden and the USA, municipalities with higher population density, in most cases, have more trees in total and a larger number of trees per capita in inventories (Kuhns et al., 2005; Östberg et al., 2018). This finding could be related to organisational and administrative differences in urban tree management between large and small LGAs (Hauer and Peterson, 2016; Östberg et al., 2018). Nonetheless, there is evidence that wealthier municipalities tend to have more trees in their inventories, compared to municipalities with less economic resources (Gerrish and Watkins, 2018; Östberg et al., 2018; Riley and Gardiner, 2020; Threlfall et al., 2022). These studies align with our findings where larger and more densely populated LGAs have higher socio-economic indices and a greater areal extent of their LGA included in urban tree

**Table 3**

Relationships between climatic, demographic, and socio-economic variables and urban forest inventory attributes in local government areas in Australia ( $n = 76$ ). Climate exposure variables refer to projected changes in 2050. LGA = local government area, IEO = Index of Education and Occupation, MTWM = maximum temperature of the warmest month, and PDQ = precipitation of the driest quarter.

| Dependent variable             | Independent variable | R <sup>2</sup> | Slope  | Statistic   | P value |
|--------------------------------|----------------------|----------------|--------|-------------|---------|
| Presence of tree inventory     | LGA area             | 0.04           | 0.004  | $z = -1.96$ | 0.04    |
| Presence of tree inventory     | Population density   | 0.06           | 0.001  | $z = 2.4$   | 0.016   |
| Areal extent of tree inventory | Population density   | 0.14           | 0.008  | $z = 2.34$  | 0.08    |
| Areal extent of tree inventory | IEO                  | 0.13           | 0.009  | $z = 2.16$  | 0.03    |
| Number of attributes collected | Number of trees      | 0.04           | 0.13   | $z = 1.72$  | 0.08    |
| IEO                            | LGA area             | 0.10           | -9.3   | $t = -3.61$ | < 0.001 |
| IEO                            | Population density   | 0.25           | 9.99   | $t = 4.23$  | < 0.001 |
| MTWM exposure                  | Population density   | 0.02           | 2.01   | $t = 72.4$  | 0.09    |
| MTWM exposure                  | LGA area             | 0.006          | -2.5   | $F = 7.71$  | 0.006   |
| MTWM exposure                  | IEO                  | 0.03           | 0.005  | $F = 4.96$  | 0.03    |
| MTWM exposure                  | PDQ exposure         | 0.14           | 0.013  | $F = 20.09$ | < 0.001 |
| MTWM exposure                  | Latitude             | 0.23           | -0.02  | $F = 34.4$  | < 0.001 |
| PDQ exposure                   | Longitude            | 0.48           | -0.4   | $F = 106.1$ | < 0.001 |
| PDQ exposure                   | LGA area             | 0.04           | -0.001 | $F = 10.4$  | 0.001   |
| PDQ exposure                   | Population size      | 0.17           | -5.7   | $F = 18.36$ | < 0.001 |
| PDQ exposure                   | Population density   | 0.07           | -4.3   | $F = 6.97$  | 0.03    |

inventories. This finding provides evidence of urban forest inequity across Australian cities. Furthermore, the number of trees in an inventory was unrelated to the number of attributes collected. LGAs with a higher number of trees in their inventories collected a similar number of attributes in their inventories compared to LGAs with fewer trees. This finding may indicate that larger LGAs have resources for data collection without compromising the number of attributes incorporated in their inventories.

Urban tree inventories can be used to study the effects of climate change within urban forests by using vulnerability assessments or climate risk metrics (e.g. Esperon-Rodriguez et al., 2022c; Hanley et al., 2021; Ordóñez and Duinker, 2014). For example, using tree inventories, a recent study in Australia showed that > 50 % of tree and shrub species planted in 22 cities are potentially at risk from climate change (Esperon-Rodriguez et al., 2019). Warmer and drier conditions can limit tree growth and performance and compromise the provision of ecosystem services (Zhang and Brack, 2021). This highlights the importance of recording tree dieback and mortality driven by climate change. Yet, among all potential causes of failure and mortality reported, heatwaves, drought and climate change were among the least frequently recorded, which might be related to a lack of long-term monitoring data to detect climate as a cause or contributing factor in tree dieback or failure (Esperon-Rodriguez et al., 2022b). Detecting and attributing tree dieback or failure to drought, heat stress and climate extremes is often complex, involving multiple factors and differing time courses and frequently a lack of diagnostic information for use by arborists compared to more direct observations of causal factors.

Current tree inventories in Australia have the potential to enable assessments of the impacts of climate change on urban forests through recording tree failure, dieback and mortality. This can be achieved by incorporating tree inventories into different mapping systems (e.g. GIS), which facilitates planning and management and is already being conducted by 55 of the 76 LGAs with inventories. These data, however, require long-term monitoring and the inclusion of site and meteorological data in the inventories. Information on tree failures, dieback and mortality can identify those species that are not performing well in a given location. This can then inform decisions on future species selection – including in other LGAs with different climate conditions – and thereby minimise economic losses (Esperon-Rodriguez et al., 2022c; McPherson et al., 2018).

Optimal species selection is crucial as it will become increasingly challenging to mitigate the effects of climate change through management actions such as irrigation (Van der Veken et al., 2008), particularly as more extreme conditions, such as heatwaves and drought, become an increasingly important feature of local climates (IPCC, 2021). More research is needed to identify climate-resilient species based on their climatic tolerance by combining ecological-functional approaches and trait-based analysis (Barradas et al., 2022; Esperon-Rodriguez et al., 2020; Farrell et al., 2022; Hirons et al., 2021; Paquette et al., 2021) to understand urban forest performance. However, for this research to be developed including as many cities as possible, tree inventories need to be available for data sharing. To date, 26 of the 76 LGAs with inventories do not provide inventory data and a further 31 will only provide data upon request, often an unreliable data-sharing strategy.

Tree inventories in Australia and elsewhere are most effective when designed to collect information on tree performance in a systematic manner and repeated over time (Roman et al., 2020; van Doorn et al., 2020). Importantly, growth data should be collected soon after planting and at regular intervals (van Doorn et al., 2020). A standardised set of metrics, including stem diameter, tree height and canopy size provide a robust basis for assessing growth performance through time (McPherson et al., 2016). These data can be used to compare species' performance in different locations. Collectively, these data provide a useful indication of how a particular species is performing in a specific urban environment.

More than 30 years ago, in 1988, John Gray wrote that “insufficient statistics were available Australia-wide to provide an accurate picture of

the urban forest estate or the rate at which it is increasing” (Gray, 1988). Unfortunately, our research shows there has not been a significant advance in the adoption and use of urban forest inventory methods over the past three decades. This study can only account for the LGAs that responded to the survey. However, it is a representation of inventories currently being managed across a large proportion of Australian LGAs. Data collected in tree inventories can aid with optimising planting outcomes and developing sustainable and resilient green cities into the future, but this will require an effort for developing urban forest strategies considering the impacts of future climate change, and currently, only half of the respondents LGAs with urban forest inventories have developed such strategies. We highlight the need for standardised metrics, as part of a long-term and dynamically updated monitoring system, along with records of planting and tree failures. Data sharing among LGAs in different locations can also help to evaluate factors contributing to good or poor growth performance and longevity of urban trees, including identifying climate resilience or vulnerability. Our results show inequalities in the establishment of urban tree inventories and provide a path forward to address deficiencies and limitations to secure resilient cities in a changing climate.

#### Author statement

MER and PDR conceived the article. MER, SAP, MGT and PDR designed the research. MER collected data. MER analysed data. MER wrote the article. All authors contributed to discussion of the content and reviewed or edited the manuscript before submission.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Acknowledgments

This study received was approved by the Human Research Ethics Committee: HREC Approval Number: H13381. We thank Amy Alessi for her contribution to the data analyses. We gratefully acknowledge Leigh Staas for her administrative support to the project. This work was funded by the Hort Frontiers Green Cities Fund, part of the Hort Frontiers strategic partnership initiative developed by Hort Innovation, with co-investment from Macquarie University, Western Sydney University and the New South Wales Department of Planning and Environment and contributions from the Australian Government. We also thank two anonymous reviewers for their comments and feedback.

#### Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ufug.2022.127769](https://doi.org/10.1016/j.ufug.2022.127769).

#### References

- Akbari, H., Pomerantz, M., Taha, H., 2001. Cool surfaces and shade trees to reduce energy use and improve air quality in urban areas. *Sol. Energy* 70, 295–310.
- Alavipanah, S., Wegmann, M., Qureshi, S., Weng, Q., Koellner, T., 2015. The role of vegetation in mitigating urban land surface temperatures: a case study of Munich, Germany during the warm season. *Sustainability* 7, 4689–4706.
- Allen, C.D., Macalady, A.K., Chenchouni, H., Bachelet, D., McDowell, N., Vennetier, M., Kitzberger, T., Rigling, A., Breshears, D.D., Hogg, E.T., 2010. A global overview of drought and heat-induced tree mortality reveals emerging climate change risks for forests. *For. Ecol. Manag.* 259, 660–684.
- Almaaitah, T., Appleby, M., Rosenblat, H., Drake, J., Joksimovic, D., 2021. The potential of Blue-Green infrastructure as a climate change adaptation strategy: a systematic literature review. *Blue-Green Syst.* 3, 223–248.
- Alvarez, I., Gallo, B., Garçon, E., Oshiro, O., 2015. Street tree inventory of Campinas, Brazil: An instrument for urban forestry management and planning. *Arboriculture & Urban. Arboricult. Urban For.* 41, 233–244.

- ASGS, 2011. Australian Statistical Geography Standard. Volume 4 - Significant Urban Areas, Urban Centres and Localities, Section of State (cat no. 1270.0.55.004). in: Statistics, C.A.B.o. (Ed.).
- BM, 2018. (Bureau of Meteorology). Special Climate Statement 66—an abnormally dry period in eastern Australia, in: Australia, C.o. (Ed.), Australia, p. 31.
- BOM and CSIRO, 2020. State of the climate. Bureau of Meteorology Australia and CSIRO. Available at (<http://www.bom.gov.au/state-of-the-climate/documents/State-of-the-climate-2020.pdf>).
- Barradas, V.L., Miranda, J.A., Esperon-Rodriguez, M., Ballinas, M., 2022. (Re) Designing Urban Parks to Maximize Urban Heat Island Mitigation by Natural Means. *Forests* 13 (7), 1143.
- Brandt, L., Lewis, A.D., Fahey, R., Scott, L., Darling, L., Swanston, C., 2016. A framework for adapting urban forests to climate change. *Environ. Sci. Policy* 66, 393–402.
- Chambers, J., Hastie, T., 1992. Linear models. Chapter 4 of statistical models in S. Wadsworth & Brooks/Cole.
- Chapman, S., Watson, J.E.M., Salazar, A., Thatcher, M., McAlpine, C.A., 2017. The impact of urbanization and climate change on urban temperatures: a systematic review. *Landsc. Ecol.* 32, 1921–1935.
- Corburn, J., 2009. Cities, climate change and urban heat island mitigation: Localising global environmental science. *Urban Stud.* 46, 413–427.
- Cresswell, I., Murphy, H., 2017. Australia state of the environment 2016: biodiversity, independent report to the Australian Government Minister for the Environment and Energy, Australian Government Department of the Environment and Energy, Canberra, ACT.
- Demuzere, M., Orru, K., Heidrich, O., Olazabal, E., Geneletti, D., Orru, H., Bhawe, A.G., Mittal, N., Feliú, E., Faehnle, M., 2014. Mitigating and adapting to climate change: Multi-functional and multi-scale assessment of green urban infrastructure. *J. Environ. Manag.* 146, 107–115.
- Di Baldassarre, G., Wanders, N., AghaKouchak, A., Kuil, L., Rangelcroft, S., Veldkamp, T. I., Garcia, M., van Oel, P.R., Breinl, K., Van Loon, A.F., 2018. Water shortages worsened by reservoir effects. *Nat. Sustain.* 1, 617.
- Esperon-Rodriguez, M., Tjoelker, M., Lenoir, J., Baumgartner, J., Beaumont, L., Nipperess, N., Power, S., Richard, B., Rymer, P., Gallagher, R., 2022. Climate change increases global risk to urban forests. *Nature Climate Change* 12 (10), 950–955.
- Esperon-Rodriguez, M., Rymer, P.D., Power, S.A., Challis, A., Marchin, R.M., Tjoelker, M. G., 2020. Functional adaptations and trait plasticity of urban trees along a climatic gradient. *Urban For. Urban Green* 54, 126771.
- Esperon-Rodriguez, M., Ordoñez, C., van Doorn, N.S., Hirons, A., Messier, C., 2022. Using climate analogues and vulnerability metrics to inform urban tree species selection in a changing climate: The case for Canadian cities. *Landsc. Urban Plan.* 228, 104578.
- Esperon-Rodriguez, M., Rymer, P.D., Power, S.A., Barton, D.N., Cariñanos, P., Dobbs, C., Eleuterio, A.A., Escobedo, F.J., Hauer, R., Hermy, M., Jahani, A., Onyekwelu, J.C., Östberg, J., Pataki, D., Randrup, T.B., Rasmussen, T., Roman, L.A., Russo, A., Shackleton, C., Solfeld, I., van Doorn, N.S., Wells, M.J., Wiström, B., Yan, P., Yang, J., Tjoelker, M.G., 2022. Assessing climate risk to support urban forests in a changing climate. *Plants People Planet* 1–13.
- Esperon-Rodriguez, M., Power, S., Tjoelker, M., Beaumont, L., Burley, H., Caballero-Rodriguez, D., Rymer, P., 2019. Assessing the vulnerability of Australia's urban forests to climate extremes. *Plants, People, Planet* 1, 1–11.
- Farrell, C., Livesley, S.J., Beaumont, L., Burley, H., Ellsworth, D., Esperon-Rodriguez, M., Fletcher, T.D., Gallagher, R.V., Ossola, A., Power, S.A., Marchin, R., Rayner, J.P., Rymer, P.D., Staas, L., Szota, C., Williams, N.S.G., Leishman, M., 2022. Can we integrate ecological approaches to improve plant selection for green infrastructure? *Urban Forestry & Urban Greening* (127732). <https://doi.org/10.1016/j.ufug.2022.127732>.
- Fick, S.E., Hijmans, R.J., 2017. WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. *Int. J. Climatol.* 37, 4302–4315.
- García-Sánchez, I., Barradas, V., de León Hill, C., Esperon-Rodriguez, M., Pérez Rosas, I., Ballinas, M., 2019. Effect of heavy metals and environmental variables on the assimilation of CO<sub>2</sub> and stomatal conductance of *Ligustrum lucidum*, an urban tree from Mexico City. *Urban For. Urban Green* 42, 72–81.
- Gerrish, E., Watkins, S.L., 2018. The relationship between urban forests and income: a meta-analysis. *Landsc. Urban Plan.* 170, 293–308.
- Gray, J., 1988. Urban forestry in Australia—the future. *Aust. For.* 51, 88–91.
- Haines, A., Kovats, R.S., Campbell-Lendrum, D., Corvalán, C., 2006. Climate change and human health: impacts, vulnerability and public health. *Public Health* 120, 585–596.
- Hanley, P.A., Arndt, S.K., Livesley, S.J., Szota, C., 2021. Relating the climate envelopes of urban tree species to their drought and thermal tolerance. *Sci. Total Environ.* 753, 142012.
- Hauer, R., Peterson, W., 2016. Municipal tree care and management in the United States: a 2014 urban & community forestry census of tree activities, Special Publication 16–1. College of Natural Resources, University of Wisconsin, Stevens Point, p. 71.
- Hilbert, D., Roman, L., Koeser, A.K., Vogt, J., Van Doorn, N.S., 2019. Urban Tree Mortality: A Literature Review. *Arboric. Urban For.* 45, 167–200.
- Hirons, A.D., Watkins, J.H.R., Baxter, T.J., Miesbauer, J.W., Male-Muñoz, A., Martin, K. W., Bassuk, N.L., Sjöman, H., 2021. Using botanic gardens and arboreta to help identify urban trees for the future. *Plants, People, Planet* 3, 182–193.
- Hoegh-Guldberg, O., Bruno, J.F., 2010. The impact of climate change on the world's marine ecosystems. *Science* 328, 1523–1528.
- Hoekstra, A.Y., Mekonnen, M.M., 2012. The water footprint of humanity. *Proc. Natl. Acad. Sci. USA* 109, 3232–3237.
- Hunt, A., Watkins, P., 2011. Climate change impacts and adaptation in cities: a review of the literature. *Clim. Change* 104, 13–49.
- IPCC, 2001. *Climate Change 2001: The Scientific Basis. Contribution of Working Group I to the Third Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.
- Italian Ministry of Ecological Transition I (I.M.o.E.T), 10/2013. in: ([https://www.mite.gov.it/sites/default/files/archivio/normativa/legge\\_14\\_01\\_2013\\_10.pdf](https://www.mite.gov.it/sites/default/files/archivio/normativa/legge_14_01_2013_10.pdf)), A.o. (Ed.), (Accessed on 1 March 2022).
- IPCC, 2021. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, in: Masson-Delmotte, V., Zhai, P., Pirani, A., Connors, S., Péan, C., Berger, S., Caud, N., Chen, Y., Goldfarb, L., Gomis, M., Huang, M., Leitzell, K., Lonnoy, E., Matthews, J., Maycock, T., Waterfield, Y., Yelekçi, O., Yu, R., Zhou, B. (Eds.).
- Khan, T., Conway, T.M., 2020. Vulnerability of common urban forest species to projected climate change and practitioners perceptions and responses. *Environ. Manag.* 65, 534–547.
- Kidder, S.Q., Essenswanger, O.M., 1995. The effect of clouds and wind on the difference in nocturnal cooling rates between urban and rural areas. *J. Appl. Meteorol.* 34, 2440–2448.
- Kuhns, M.R., Lee, B., Reiter, D.K., 2005. Characteristics of urban forestry programs in Utah. *U.S. J. Arboric.* 31, 285.
- Legendre, P., Legendre, L.F., 2012. *Numerical Ecology*, third edition. Elsevier, Oxford, UK.
- Lenoir, J., Bertrand, R., Comte, L., Bourgeaud, L., Hattab, T., Muriene, J., Grenouillet, G., 2020. Species better track climate warming in the oceans than on land. *Nat. Ecol. Evol.* 4, 1044–1059.
- Livesley, S., McPherson, E.G., Calafapietra, C., 2016. The urban forest and ecosystem services: impacts on urban water, heat, and pollution cycles at the tree, street, and city scale. *J. Environ. Qual.* 45, 119–124.
- Ma, B., Hauer, R.J., Östberg, J., Koeser, A.K., Wei, H., Xu, C., 2021. A global basis of urban tree inventories: What comes first the inventory or the program. *Urban For. Urban Green* 60, 127087.
- Manoli, G., Faticchi, S., Schläpfer, M., Yu, K., Crowther, T.W., Meili, N., Burlando, P., Katul, G.G., Bou-Zeid, E., 2019. Magnitude of urban heat islands largely explained by climate and population. *Nature* 573, 55–60.
- Marchin, R.M., Esperon-Rodriguez, M., Tjoelker, M.G., Ellsworth, D.S., 2022. Crown dieback and mortality of urban trees linked to heatwaves during extreme drought. *Sci. Total Environ.* 850, 157915.
- McPherson, E.G., Berry, A.M., van Doorn, N.S., 2018. Performance testing to identify climate-ready trees. *Urban For. Urban Green* 29, 28–39.
- McPherson, E.G., van Doorn, N.S., Peper, P.J., 2016. *Urban tree database and allometric equations*, Gen. Tech. Rep. PSW-GTR-253. Albany, CA: US Department of Agriculture, Forest Service, Pacific Southwest Research Station, p. 86.
- Miller, R.W., Hauer, R.J., Werner, L.P., 2015. *Urban Forestry: Planning and Managing Urban Greenspaces*, 3rd edition. Waveland Press, Inc, Long Grove, Illinois.
- Morgenroth, J., Östberg, J., van den Bosch, C.K., Nielsen, A.B., Hauer, R., Sjöman, H., Chen, W., Jansson, M., 2016. Urban tree diversity—Taking stock and looking ahead. *Urban For. Urban Green* 15, 1–5.
- Nielsen, A.B., Östberg, J., Delshamar, T., 2014. Review of urban tree inventory methods used to collect data at single-tree level. *Arboric. Urban For.* 40, 96–111.
- Nowak, D.J., Greenfield, E.J., 2018. Declining urban and community tree cover in the United States. *Urban For. Urban Green* 32, 32–55.
- Oksanen, G., Blanchet, F., Friendly, M., Kindt, R., Legendre, P., McGinn, D., Minchin, P., O'Hara, R., Simpson, G., Solymos, P., Stevens, H., Szoecs, E., Wagner, H., 2019. *Vegan. Community Ecology Package*.
- Ordóñez, C., Duinker, P., 2014. Assessing the vulnerability of urban forests to climate change. *Environ. Rev.* 22, 311–321.
- Östberg, J., Wiström, B., Randrup, T.B., 2018. The state and use of municipal tree inventories in Swedish municipalities—results from a national survey. *Urban Ecosyst.* 21, 467–477.
- Paquette, A., Sousa-Silva, R., Maure, F., Cameron, E., Belluau, M., Messier, C., 2021. Praise for diversity: a functional approach to reduce risks in urban forests. *Urban For. Urban Gree.* 62, 127157.
- Páscoa, P., Gouveia, C.M., Russo, A., Ribeiro, A.F.S., 2022. Summer hot extremes and antecedent drought conditions in Australia. *International Journal of Climatology* n/a.
- R Core Team, 2021. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria.
- Ramage, B.S., Roman, L.A., Dukes, J.S., 2013. Relationships between urban tree communities and the biomes in which they reside. *Appl. Veg. Sci.* 16, 8–20.
- Riley, C.B., Gardiner, M.M., 2020. Examining the distributional equity of urban tree canopy cover and ecosystem services across United States cities. *PLoS One* 15, e0228499.
- Roman, L.A., Battles, J.J., McBride, J.R., 2014. Determinants of establishment survival for residential trees in Sacramento County, CA. *Landsc. Urban Plan.* 129, 22–31.
- Roman, L.A., Walker, L.A., Martineau, C.M., Muffy, D.J., MacQueen, S.A., Harris, W., 2015. Stewardship matters: Case studies in establishment success of urban trees. *Urban For. Urban Green* 14, 1174–1182.
- Roman, L.A., van Doorn, N.S., McPherson, E.G., Scharenbroch, B.C., Henning, J.G., Östberg, J.P., Mueller, L.S., Koeser, A.K., Mills, J.R., Hallett, R.A., 2020. *Urban Tree Monitoring: A Field Guide*. Gen. Tech. Rep. NRS-194, 194. US Department of Agriculture, Forest Service, Northern Research Station, Madison, WI, pp. 1–48.
- Roman, L.A., Fristensky, J.P., Lundgren, R.E., Cerwinka, C.E., Lubar, J.E., 2022. Construction and proactive management led to tree removals on an urban college campus. *Forests* 13, 871.
- Sanusi, R., Livesley, S.J., 2020. London Plane trees (*Platanus x acerifolia*) before, during and after a heatwave: Losing leaves means less cooling benefit. *Urban For. Urban Green* 54, 126746.



- Scholze, M., Knorr, W., Arnell, N.W., Prentice, I.C., 2006. A climate-change risk analysis for world ecosystems. *Proc. Natl. Acad. Sci. USA* 103, 13116–13120.
- Teodorescu, G., 2010. Climate change impact on urban ecosystems and sustainable development of cities in Romania. *WSEAS Trans. Environ. Dev.* 6, 103–112.
- Threlfall, C.G., Gunn, L.D., Davern, M., Kendal, D., 2022. Beyond the luxury effect: Individual and structural drivers lead to ‘urban forest inequity’ in public street trees in Melbourne, Australia. *Landsc. Urban Plan.* 218, 104311.
- Van der Veken, S., Hermy, M., Vellend, M., Knapen, A., Verheyen, K., 2008. Garden plants get a head start on climate change. *Front. Ecol. Environ.* 6, 212–216.
- van Doorn, N.S., Roman, L.A., McPherson, E.G., Scharenbroch, B.C., Henning, J.G., Östberg, J.P., Mueller, L.S., Koeser, A.K., Mills, J.R., Hallet, R.A., 2020. Urban Tree Monitoring: A Resource Guide. Gen. Tech. Rep. PSW-GTR-266, 132. US Department of Agriculture, Forest Service, Pacific Southwest Research Station, Albany, CA, p. 266.
- Vogt, J., Gillner, S., Hofmann, M., Tharang, A., Dettmann, S., Gerstenberg, T., Schmidt, C., Gebauer, H., Van de Riet, K., Berger, U., 2017. Citree: a database supporting tree selection for urban areas in temperate climate. *Landsc. Urban Plan.* 157, 14–25.
- Young, R.F., McPherson, E.G., 2013. Governing metropolitan green infrastructure in the United States. *Landsc. Urban Plan.* 109, 67–75.
- Zhang, B., Brack, C.L., 2021. Urban forest responses to climate change: a case study in Canberra. *Urban For. Urban Green* 57, 126910.